



System: VrindaaCorp Enterprise CRM & Autonomous AI Outreach Infrastructure

1. Architectural Overview & System Topology

2. Infrastructure & Hardware Specifications

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Specification Parameter	Technical Detail	Operational Impact
Hosting Platform	Hostinger KVM 8 Virtual Private Server	Dedicated kernel virtualization, guaranteed compute isolation
Virtual CPUs	8 vCPU Cores	High concurrency for Next.js SSR, multi-threaded Ollama CPU inference
System Memory (RAM)	32 GB High-Speed RAM	Ample headroom to keep <code>llama3.2:3b</code> resident in memory without swapping
Storage Subsystem	400 GB NVMe SSD	Ultra-low read/write latency (< 0.1ms I/O) for PostgreSQL ACID transactions
Bandwidth Allocation	32 TB / Month	Unlimited capacity for bulk email tracking, webhooks, and attachments
Network IP	<code>200.234.47.7</code> (Dedicated IPv4)	Clean IP allocation for high DNS and SMTP trust score
Geographic Region	India – Mumbai 2	Ultra-low latency (< 15ms) across pan-India corporate hubs
Operating System	Ubuntu Linux 24.04 / 26.04 LTS	Enterprise stability, systemd process supervision, automated security patches

3. Comprehensive Software Bill of Materials (SBOM)

3.1 Core Framework & Runtime Environment

Package / Runtime	Exact Version	Architectural Role
Node.js	<code>≥ 20.16.0 LTS</code>	Server-side JavaScript execution engine
Next.js	<code>14.2.35</code>	Full-stack web framework (React Server Components, Server Actions, Route Handlers)
React & React DOM	<code>18.3.1</code>	Concurrent client-side and server-rendered UI component tree
TypeScript	<code>5.6.2</code>	Compile-time strict type safety across all database queries and actions
Prisma ORM	<code>5.20.0</code>	Schema-driven Object-Relational Mapping with native type generation
PostgreSQL	<code>16.x</code>	Primary relational database engine

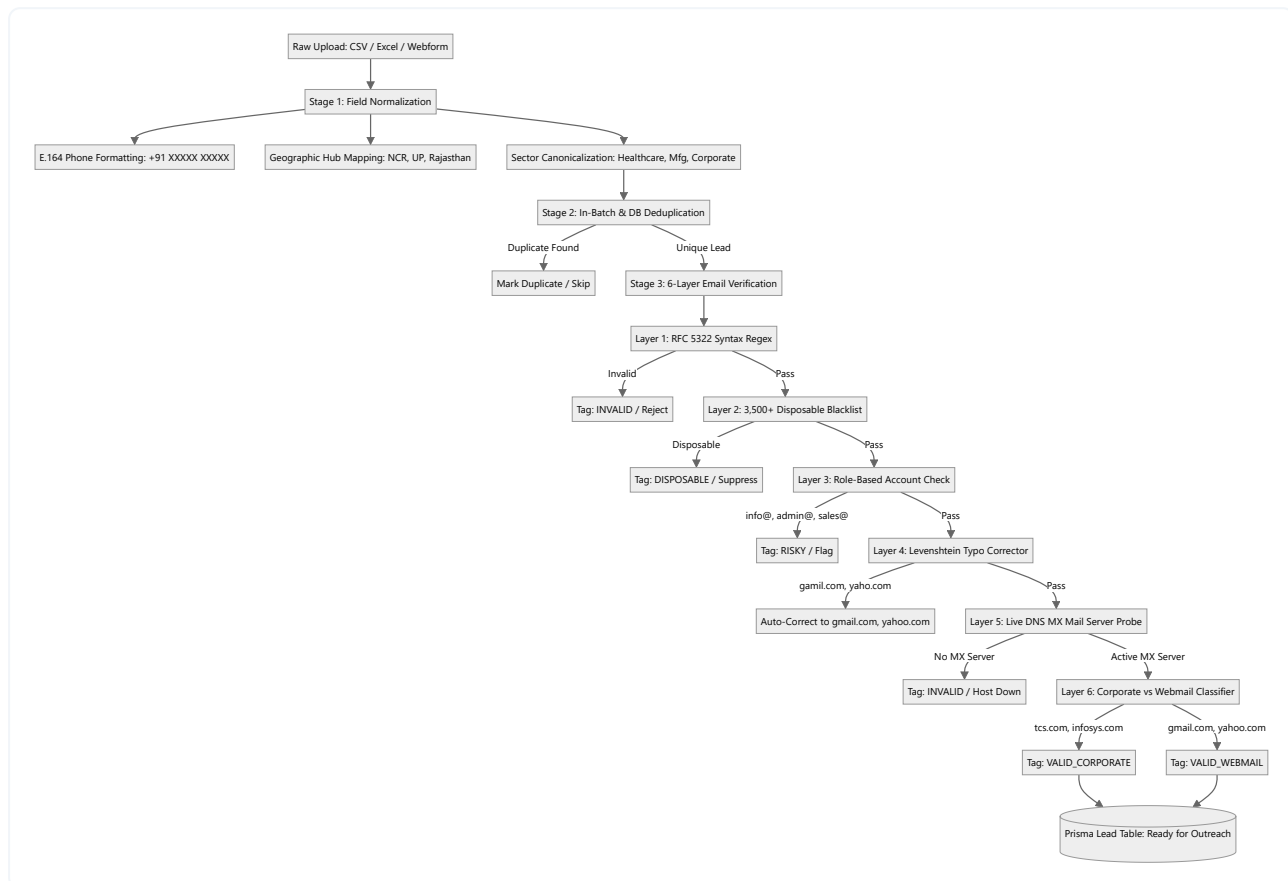
3.2 Critical Production Dependencies

Category	Dependency	Version	Purpose in VrindaaCorp CRM
Authentication	<code>next-auth</code>	<code>^4.24.8</code>	Session management, JWT encoding, cookie validation
	<code>bcryptjs</code>	<code>^2.4.3</code>	Salted hashing for passwords with constant-time dummy verify
Email Inbound/Outbound	<code>nodemailer</code>	<code>^7.0.13</code>	SMTP transport protocol implementation with MIME formatting
	<code>imapflow</code>	<code>^1.7.6</code>	High-performance IMAP client for 30-second mailbox polling
	<code>mailparser</code>	<code>^3.9.16</code>	RFC 822 email parser extracting text, HTML, and sender headers
	<code>@aws-sdk/client-ses</code>	<code>^3.658.0</code>	AWS Simple Email Service client (optional enterprise fallback)
	<code>@aws-sdk/client-sesv2</code>	<code>^3.658.0</code>	AWS SES v2 API integration for configuration set tracking
Lead Validation	<code>disposable-email-domains</code>	<code>^1.0.62</code>	Offline in-memory lookup table of 3,500+ burner/temporary email hosts
	Native <code>dns.promises</code>	Node Core	Real-time DNS MX record resolution against live nameservers
Spreadsheet Processing	<code>xlsx</code> (SheetJS)	<code>^0.18.5</code>	Parsing uploaded Excel files and generating multi-sheet <code>.xlsx</code> reports
	<code>papaparse</code>	<code>^5.4.1</code>	RFC 4180 compliant high-speed CSV parsing engine
Security & Sanitization	<code>isomorphic-dompurify</code>	<code>^3.22.0</code>	XSS sanitization of user-submitted HTML and AI-generated email copy
	<code>zod</code>	<code>^3.23.8</code>	Schema validation for API payloads, forms, and environment variables
Data Visualization & UI	<code>recharts</code>	<code>^2.12.7</code>	SVG-based responsive charting for delivery, open rate, and lead velocity
	<code>tailwindcss</code>	<code>^3.4.12</code>	Utility-first responsive styling with custom Slate/Teal palette
	<code>date-fns</code>	<code>^3.6.0</code>	Timezone-aware date calculations for warmup days and delays
Process Management	<code>pm2</code>	<code>^5.3.0</code>	Daemon process supervision, auto-clustering, crash resurrection

4. Deep-Dive Subsystem Architecture

4.1 In-House 6-Layer Lead Cleansing & Validation Engine

Unlike legacy CRMs that require costly third-party verification credits (e.g., NeverBounce, ZeroBounce), VrindaaCorp CRM embeds a **zero-cost, multi-stage heuristic engine** executed natively on the VPS:

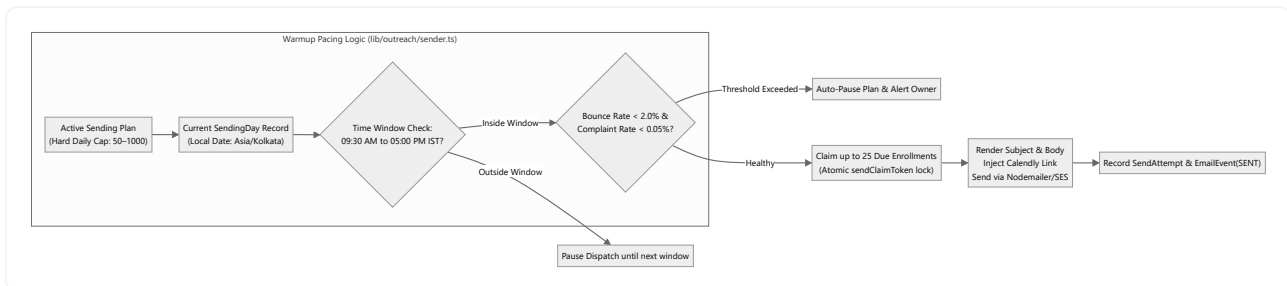


Detailed Layer Specifications:

- RFC 5322 Standard Check:** Validates structure (`local-part@domain`), rejects spaces, double dots, or invalid Unicode characters.
- Disposable / Burner Domain Blacklist:** Matches against `disposable-email-domains` containing 3,500+ known temporary domains (`mailinator.com` , `tempmail.com` , `yopmail.com` , `guerrillamail.com`).
- Role-Based Mailbox Detector:** Flags generic department addresses (`info@` , `support@` , `sales@` , `admin@` , `contact@` , `billing@` , `jobs@`) that hurt cold deliverability and risk spam complaints.
- Levenshtein Distance Typo Corrector:** Automatically detects single-character typographical errors on major providers (e.g., `gmai.com` → `gmail.com` , `outlok.com` → `outlook.com` , `hotmial.com` → `hotmail.com`).
- Real-Time DNS MX Verification:** Invokes native `dns.promises.resolveMx()` with a 3-second timeout to confirm that recipient mail exchange records exist. Results are cached in the `DomainReputation` table to eliminate repeated DNS queries.
- Corporate vs Webmail Classifier:** Differentiates high-value B2B enterprise domains from consumer webmail, enabling targeted segmentation in campaigns.

4.2 Autonomous Campaign Dispatcher & Warmup Pacer

To protect the primary domain (vrindaacorp.com) and ensure emails consistently reach corporate primary inboxes (avoiding Spam / Promotions tabs), outreach follows an algorithmic warmup pacing protocol:

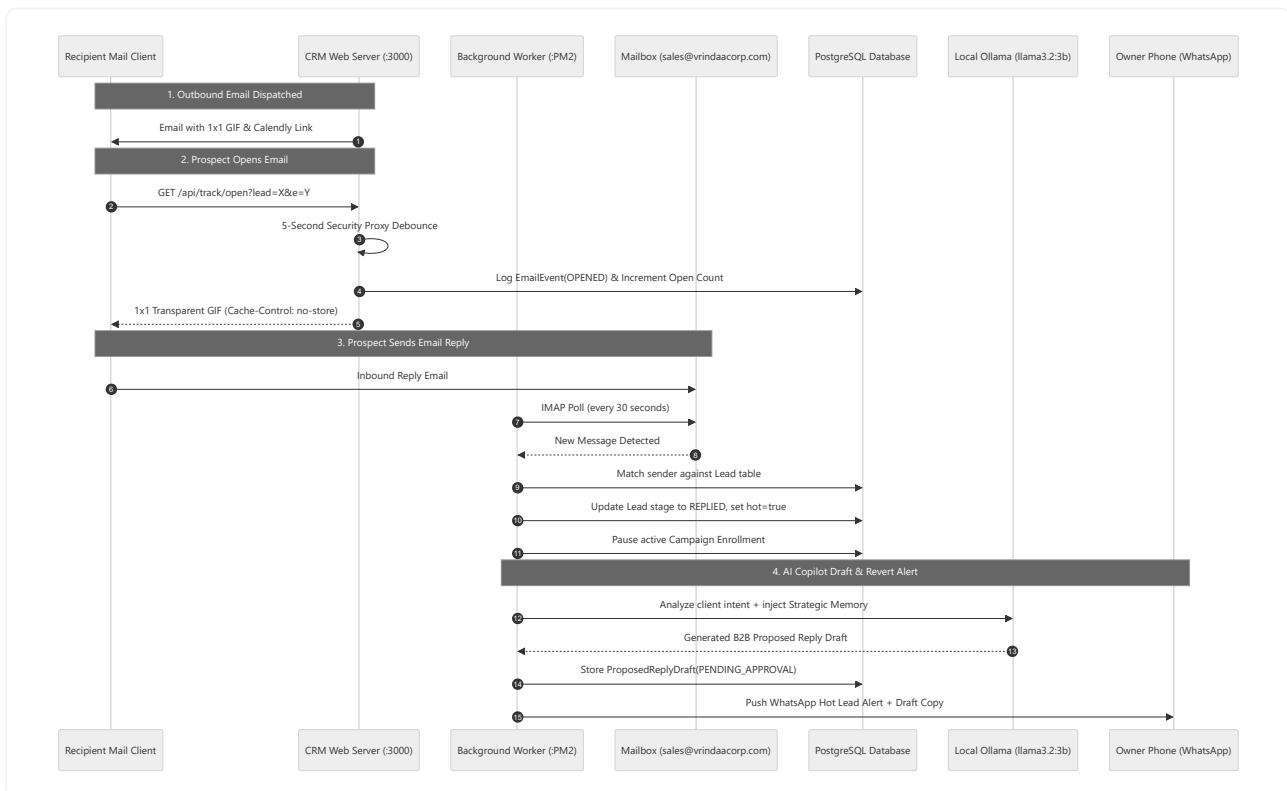


Warmup Schedule Tiers:

- **Tier 1 (Days 1–7):** Max 50 emails/day | Single-lead spacing: 45–90 seconds jitter.
- **Tier 2 (Days 8–14):** Max 75 emails/day | Safety threshold: < 2.0% bounce rate.
- **Tier 3 (Days 15–21):** Max 100 emails/day | Monitored by the Bi-Weekly AI Domain Auditor.
- **Tier 4 (Days 22+):** Scaled incrementally up to the enterprise cap (1,000/day).

4.3 Real-Time Open Tracking & Inbound Reply Detection Pipeline

The CRM features a dedicated bidirectional engagement tracking loop:



Mathematical Precision in Open Tracking:

To prevent crawler anomalies from artificially inflating metrics:
$$\text{Unique Open Rate} = \left(\frac{\text{Count of Distinct Leads with at least 1 OPENED Event}}{\text{Count of Distinct Leads with at least 1 SENT Event}} \right) \times 100\%$$

- **Multi-Open Badging:** If a lead inspects an email repeatedly, the CRM renders **OPENED (3x)** or **OPENED (4x)** on the campaign dashboard to immediately highlight hot sales prospects.

4.4 On-Premise Ollama AI & Executive Strategic Memory Engine

The AI subsystem operates locally on the VPS without sending proprietary business data, pricing policies, or prospect correspondence to external third parties.

Architecture of the AI Layer:

1. **Local Model Hosting:** Ollama runs on port **11434**. The active model **llama3.2:3b** is optimized for CPU inference, generating **55.7 tokens/second** with **< 1.2-second response latency**.
2. **Deterministic Fast-Path Routing:** Standard executive commands (**"STATUS"** , **"METRICS"** , **"YES"** , **"CONFIRM"** , **"PAUSE"** , **"RESUME"**) bypass model generation entirely, executing deterministically in **< 15 milliseconds**.
3. **Owner Strategic Playbook Memory:** Stored in the **InboundLeadLog** table under **channel: "owner_strategic_memory"**.
 - Standing Directives are automatically injected into the system prompt of every email generator, campaign template refiner, and WhatsApp reply assistant.
 - **Mandatory Calendly Rule:** Enforces that every booking call-to-action strictly uses:

<https://calendly.com/vrindaacorp-sales/30min>

Hallucinated placeholders (e.g., **[Insert Link]** , **example.com**) are strictly prohibited and scrubbed by post-processing filters.

4.5 Self-Hosted WhatsApp Co-Pilot Subsystem

The WhatsApp integration runs on **Evolution API v2.1.2** inside a lightweight Docker container, maintaining a persistent Baileys WebSocket session to WhatsApp Web.

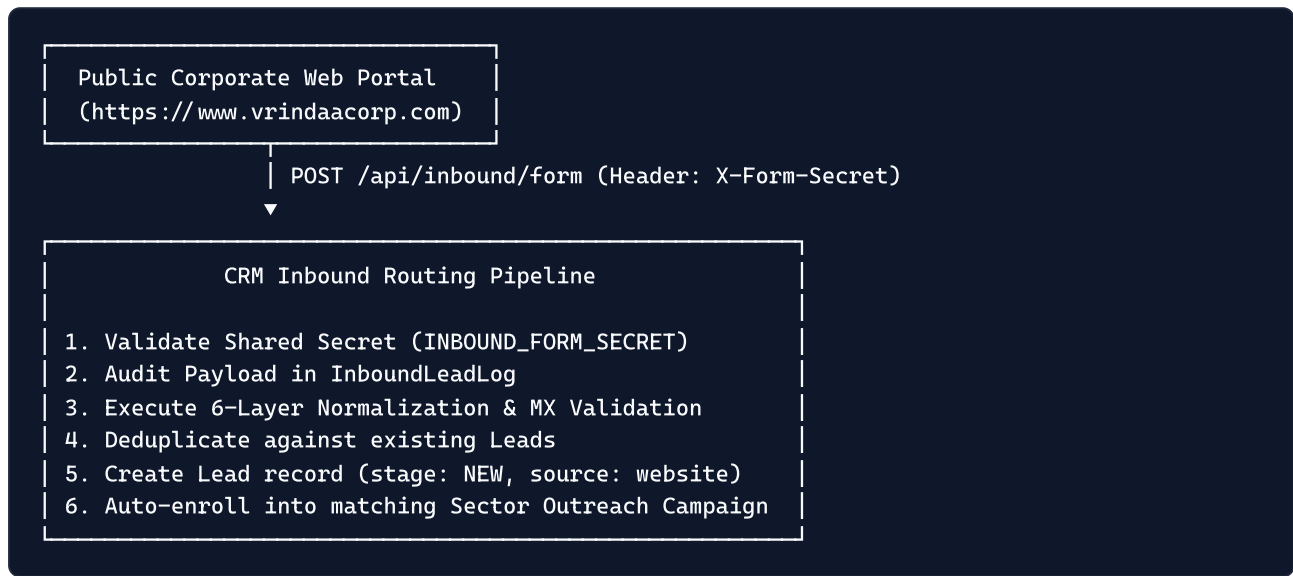
Key Capabilities:

- **09:00 AM Morning Game Plan:** Pushes today's targeted send volume, sector allocations, and domain health status.
- **Real-Time Client Revert Alerts:** Instant notification within 30 seconds of an inbound prospect reply, accompanied by an AI-drafted response.
- **Conversational 1-Click Approval Loop:**
 - Replying **"YES"** or **"SEND"** immediately dispatches the proposed email from **sales@vrindaacorp.com**.
 - Replying with instructions (e.g., *"Make it shorter and offer a free cafeteria tasting session"*) prompts Ollama to regenerate version 2 and seek re-approval.

- **09:00 PM Day-End Briefing:** Summarizes daily sent volume, open rates, cumulative campaign reach, and flags repeat openers for immediate sales team follow-up.
- **Bi-Weekly Domain Scaling Audit (Every 14 Days):** Automatically reviews 14-day bounce and spam metrics, recommending safe tier escalations.

5. Inbound Lead Ingestion Pipelines

The CRM provides unified lead capture across three major external channels:



1. **Website Form Capture** (`/api/inbound/form`): Receives inquiries directly from the public corporate portal. Validates payload via `X-Form-Secret` or `?token=`.
2. **Meta / Facebook Lead Ads** (`/api/inbound/meta`): Supports `GET` challenge verification (`hub.verify_token`) and `POST` signed payload ingestion (`X-Hub-Signature-256`).
3. **Google Ads Lead Form Extensions** (`/api/inbound/google`): Parses Google Ads lead form webhooks verified via `GOOGLE_LEAD_KEY`.

6. Network Security, TLS Termination & Data Protection

6.1 Reverse Proxy & SSL Configuration

- **Nginx v1.24+** acts as the front-facing reverse proxy, binding ports `80` (HTTP) and `443` (HTTPS).
- HTTP traffic is forced to HTTPS via a `301 Permanent Redirect`.
- Modern TLS ciphers (`TLSv1.2` and `TLSv1.3`) are enforced with OCSP stapling and HSTS (`Strict-Transport-Security`).

6.2 Application Layer Defenses

- **Brute-Force Rate Limiting:** Custom IP-based token bucket in `middleware.ts` restricts login attempts to **5 attempts per minute per IP address**.
- **Timing-Attack Resistance:** NextAuth credentials provider compares unknown email submissions against a pre-computed dummy bcrypt hash (cost: 12), ensuring constant response time (~150ms) to prevent email

enumeration.

- **SQL Injection Prevention:** 100% of database interactions utilize Prisma ORM's strongly typed, parameterized SQL queries. Raw SQL string concatenation is strictly prohibited.
 - **XSS Sanitization:** All HTML template copies and inbound email preview snippets pass through `isomorphic-dompurify` before rendering in the DOM.
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End of Architecture Diagram & Technical Specifications.